Practical - 6

Aim:

Create and utilize the user defined functions.

Q1: Write a program to perform addition of two numbers using user defined function.

Code:

```
def add(a, b):
  return a + b

a = int(input("Enter A: "))
b = int(input("Enter B: "))
print("Sum is:", add(a, b))
```

Output:

```
Enter A: 5
Enter B: 6
Sum is: 11
```

Q2: Write a program to display all the prime numbers between 1 to n using function.

Code:

```python
def isPrime(n):
  for i in range(2, (n // 2) + 1):
    if n % i == 0:
      return False
  return True

def primeList(n):
  primes = []
  for i in range(1, n):
    if isPrime(i):
      primes.append(i)
  return primes

n = int(input("Enter Number That You Want Prime Till: "))
print(primeList(n))
```

Output:

```
Enter Number That You Want Prime Till: 50
[1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
```

Q3: Write a user defined function to sort a List.

Code:

```
def bubble(a):
  for i in range(len(a)):
    for j in range(i + 1, len(a)):
      if a[i] > a[j]:
        a[i], a[j] = a[j], a[i]
  return a

a = [5, 4, 3, 2, 3, 2, 1]

print(f"""
Unsorted List: {a}
Sorted List: {bubble(a)}
""")
```

Output:

```
Unsorted List: [5, 4, 3, 2, 3, 2, 1]
Sorted List: [1, 2, 2, 3, 3, 4, 5]
```

Q4: Write a function to find the minimum and maximum value from argument list & return both minimum & maximum in tuple form.

Code:

```
def min_max(a):
    minimum = a[0]
    maximum = a[0]

    for val in a:
        if val < minimum:
            minimum = val
        elif val > maximum:
            maximum = val

    return (minimum, maximum)

a = [5, 4, 3, 2, 3, 2, 1]
print(f"""
List: {a}
Min & Max: {min_max(a)}
""")
```

Output:

```
List: [5, 4, 3, 2, 3, 2, 1]
Min & Max: (1, 5)
```

Q5: Write a function to add two lists of the same length term-by-term & return new list

Eg.: A=listAdd([1,2,3],[1,2,3])

print (A) Will print [2,4,6].

Code:

```
def listAdd(A, B):
  if len(A) != len(B):
    return None
  sum = [0 for i in range(len(A))]
  for i in range(len(A)):
    sum[i] = A[i] + B[i]
  return sum

a = [5, 4, 3]
b = [5, 2, 1]

print(f"""
list1: {a}
list2: {b}
Added List: {listAdd(a, b)}
""")
```

Output:

```
list1: [5, 4, 3]

list2: [5, 2, 1]

Added List: [10, 6, 4]
```

Q6: WAP a function called powers(n) that prints out the first 5 powers of a given number.

Eg. >>> powers(6)

The first 5 powers of 6 are: 1 6 36 216 1296

Code:

```
def powers(n):
  print(f"The first 5 powers of {n} are: ", end='')
  for i in range(5):
    print((n ** i), end=' ')

powers(6)
```

Output:

```
The first 5 powers of 6 are: 1 6 36 216 1296
```

Q7: Write a Python program to filter a list of integers using lambda function. The program should filter even numbers from a given list.

Code:

```
a = [5, 4, 3, 2, 3, 2, 1]
ans = list(filter(lambda x: x % 2 != 0, a))
print(f"""
list: {a}
Even Number are Filtered: {ans}
""")
```

Output:

```
list: [5, 4, 3, 2, 3, 2, 1]
Even Number are Filtered: [5, 3, 3, 1]
```

Q8: Write a Python program to sort a list of dictionaries using lambda function. The program should sort the list based on the 'age' key in each dictionary.

Code:

```
profiles = [
    {'name':"Vatsal", 'age':21},
    {'name':"Honey", 'age':22},
    {'name':"Subodh", 'age':18}
]

def sortAge(A):
  for i in range(len(A)):
    for j in range(i+1, len(A)):
      if A[i]['age'] > A[j]['age']:
        A[i], A[j] = A[j], A[i]
  return A

print(sortAge(profiles))
```

Output:

```
[{'name': 'Subodh', 'age': 18}, {'name': 'Vatsal', 'age': 21},
{'name': 'Honey', 'age': 22}]
```

Q9: Write a Python program to execute a function on each element of a list using lambda function. The program should calculate square of all numbers.

Code:

```
a = [1, 2, 3, 4, 5, 6]
ans = list(map(lambda x : x ** 2, a))
print(f"""
list: {a}
Squared List: {ans}
""")
```

Output:

```
list: [1, 2, 3, 4, 5, 6]
Squared List: [1, 4, 9, 16, 25, 36]
```

Practice Exercise:

1. Write a program that implements both regular and lambda functions for a simple calculator:

- Create normal functions for add, subtract, multiply, divide
- Create equivalent lambda functions
- Take two numbers and operation choice from user
- Compare results from both function types
- Example: calc(10, 5, 'multiply') should return 50

Code:

```
# Regular Functions
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Error: Division by zero"
    return a / b


# Lambda Functions
lambda_add = lambda a, b: a + b
lambda_subtract = lambda a, b: a - b
lambda_multiply = lambda a, b: a * b
lambda_divide = lambda a, b: "Error: Division by zero" if b == 0
else a / b


# Calculator Function
def calc(a, b, operation):
    if operation == 'add':
        regular_result = add(a, b)
        lambda_result = lambda_add(a, b)

    elif operation == 'subtract':
        regular_result = subtract(a, b)
        lambda_result = lambda_subtract(a, b)

    elif operation == 'multiply':
        regular_result = multiply(a, b)
        lambda_result = lambda_multiply(a, b)
```

```
    elif operation == 'divide':
        regular_result = divide(a, b)
        lambda_result = lambda_divide(a, b)

    else:
        return "Invalid operation"

    return regular_result, lambda_result


# Taking input from user
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
operation = input("Enter operation (add, subtract, multiply,
divide): ").lower()

result = calc(num1, num2, operation)

print("Regular Function Result:", result[0])
print("Lambda Function Result:", result[1])
```

Output:

```
Enter first number: 66
Enter second number: 92
Enter operation (add, subtract, multiply, divide): multiply
Regular Function Result: 6072.0
Lambda Function Result: 6072.0
```

2. Create a program that works with lists using various function types:

- Write a function to find all perfect numbers up to n
- Create a lambda function to filter numbers divisible by both 2 and 3
- Write a function to return [sum, average, product] of list elements
- Example: list_stats([1,2,3,4]) should return [10, 2.5, 24]

Code:

```
from functools import reduce

# Task 1
def isPerfect(n):
  divisors = []
  for i in range(1, int((n/2) + 1)):
    if n % i == 0:
      divisors.append(i)
  if sum(divisors) == n:
    return True
  else:
    return False

def perfectTillN(n):
  perfects = []
  for i in range(1, n):
      if  isPerfect(i):
        perfects.append(i)
  return perfects

n = int(input("Enter Number You Want to Get Perfect Number Till:"))
print(f"Perfect Number Till {n}",perfectTillN(2000))

# Task 2
list1 = [i for i in range(1, 30)]
print("Number Divisible by 2 Are Filtered", list(filter(lambda x :
x % 2 != 0, list1)))
```

```python
print("Number Divisible by 3 Are Filtered", list(filter(lambda x :
x % 3 != 0, list1)))

# Task 3
list1 = [1, 2, 3, 4]
def analyze_list(l):
  sum = reduce(lambda x, y : x + y, l)
  average = sum / len(l)
  product = reduce(lambda x, y: x * y, l)
  return [sum, average, product]

print("analyze_list Output:", analyze_list(list1))
```

Output:

```
Enter Number You Want to Get Perfect Number Till:500
Perfect Number Till 500 [6, 28, 496]
Number Divisible by 2 Are Filtered [1, 3, 5, 7, 9, 11, 13, 15, 17,
19, 21, 23, 25, 27, 29]
Number Divisible by 3 Are Filtered [1, 2, 4, 5, 7, 8, 10, 11, 13,
14, 16, 17, 19, 20, 22, 23, 25, 26, 28, 29]
analyze_list Output: [10, 2.5, 24]
```

3. Write a program for student grade analysis using functions:

- Create a regular function to calculate percentage from marks list
- Use lambda to filter students scoring above 75%
- Function to return tuple of (highest_scorer_name, lowest_scorer_name)
- Example: analyze_grades({'John':[80,85,90], 'Mike':[70,65,75]})

Code:

```
def student_persentage(l):
  return sum(l)/len(l)

def transform_marks_array_in_persentage(d_marks):
  transformed_d = {}
  for key, val in d_marks.items():
    transformed_d[key] = student_persentage(val)
  return transformed_d

def top_bottom_grades(d_marks):
  student_percentages= transform_marks_array_in_persentage(d_marks)

  if not student_percentages:
    return (None, None) # Handle empty dictionary case

  highest_scorer_name = max(student_percentages,
key=student_percentages.get)
  lowest_scorer_name = min(student_percentages,
key=student_percentages.get)

  return (highest_scorer_name, lowest_scorer_name)

def analyze_grades(d):
  percentages_dict = transform_marks_array_in_persentage(d)
  print(f"Percentage Dict: {percentages_dict}")

  d1 = dict(filter(lambda x: x[1] >= 75, percentages_dict.items()))
  print(f"Filtered Dict: {d1}")

  print(f"Highest & Lowest Scorer Name: {top_bottom_grades(d)}")

d_original = {'John':[80,85,90], 'Mike':[70,65,75]}

analyze_grades(d_original)
```

Output:

```
Percentage Dict: {'John': 85.0, 'Mike': 70.0}

Filtered Dict: {'John': 85.0}

Highest & Lowest Scorer Name: ('John', 'Mike')
```

4. Create a program with following list manipulation functions:

- Function to merge two lists alternately
- Lambda to filter elements present in both lists
- Function to return every nth element from merged list
- Example: merge_lists([1,2,3], [a,b,c]) should return [1,a,2,b,3,c]

Code:

```
def merge_lists(a, b):
  listIdx = 0
  c = []
  for i in range(0, len(a)):
      c.append(a[listIdx])
      c.append(b[listIdx])
      listIdx += 1
  return c

a = [1,2,3]
b = ['a','b','c']

merged_list = merge_lists(a, b)
print(f"Merged List: {merged_list}")

a = [1, 2, 3, 4, 5]
b = [1, 2, 1, 2]
print(f"Filtered list: {list(filter(lambda x: x in b, a))}")

def nth_elements(a, n=1):
  print(f"Every {n} element of array: ", end='')
  for i in range(0, len(a), n):
      print(a[i],end=' ')

nth_elements(a, 2)
```

Output:

```
Merged List: [1, 'a', 2, 'b', 3, 'c']

Filtered list: [1, 2]

Every 2 element of array: 1 3 5
```

5. Develop a program for text analysis using functions:

- Function to count words, characters, spaces
- Lambda to filter words longer than n characters
- Function to return tuple of (most_frequent_word, frequency)
- Example: analyze_text('hello world hello python') should return [3, 16, 2]

Code:

```
def count_words(s):
  return len(s.split())

def count_total_characters(s):
  return len(s)

def count_space(s):
  c = 0
  for i in s:
    if i == ' ':
      c += 1
  return c

def count_characters(s):
  return count_total_characters(s) - count_space(s)

def analyze_text(s):
  return [count_words(s), count_characters(s), count_space(s)]

string = input("Enter String: ")
analyzed_str = analyze_text(string)

print(f"""
Words: {analyzed_str[0]}
Characters: {analyzed_str[1]}
Spaces: {analyzed_str[2]}
""")

n = int(input("Enter n so filter words longer than n characters: "))
# n = 1
s = " ".join(list(filter(lambda x: len(x) < n, string.split())))
print(s,"\n")

def get_most_frequent_word(s):
  d = {}
  for i in s.split():
    if d.get(i) == None:
      d[i] = 1
    else:
      d[i] += 1
```

```
  max = s.split()[0]
  for key, val in d.items():
    if val > d[max]:
      max = key
  return (max, d[max])

print(f"Most Frequent Word: {(get_most_frequent_word(string))}")
```

Output:

```
Enter String: Hii My Name is Vatsal Patel. I am Doing Computer
Engineering From Ganpat University. I am Fine.

Words: 17
Characters: 79
Spaces: 16

Enter n so filter words longer than n characters: 4
Hii My is I am I am

Most Frequent Word: ('I', 2)
```